

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0831

Smithville 138 kV
May 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0831. The Interconnection Customer proposes a 127.7 MW natural gas combined cycle facility interconnecting at the Smithville 138 kV substation in the Pennsylvania area. The request was received on July 20, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 272 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,782,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SHAW - WFEC GENCO LLC 138 kV	P.T.S - MEDORA 345 kV L	150.7	168	6.2
WFEC GENCO LLC - KEMPSTER 138 kV	MEDORA - P.T.S 345 kV	148.6	172	6.2
HALF MOON - SMITHVILLE 138 kV line	SIMPSON STUART TAP - KEMPSTER 138 kV	142.3	221	65.7
KEMPSTER - SIMPSON STUART TAP 138 kV	JOHN DAY - TUCO 138 kV	133.3	325	11.9
SHAW - WINEAGLE 138 kV	P.T.S - MEDORA 345 kV L	117.1	204	6.2
COFFEYVILLE - FAP 138 kV line	WFEC GENCO LLC - KEMPSTER SUBSTATION 138 kV L	104.7	161	6.5
TUCO - JOHN DAY SUBSTATION 138 kV	SIMPSON STUART TAP - KEMPSTER 138 kV	103.5	326	17.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WFEC GENCO LLC SUBSTATION - SHAW 138 kV	Rebuild line with higher-capacity conductor	\$3,228,000	100% (\$3,228,000)
WFEC GENCO LLC - KEMPSTER 138 kV	Rebuild line with higher-capacity conductor	\$6,367,000	100% (\$6,367,000)
HALF MOON - SMITHVILLE 138 kV line	Rebuild line with higher-capacity conductor	\$598,000	91% (\$544,000)
SIMPSON STUART TAP - KEMPSTER 138 kV	Rebuild line with higher-capacity conductor	\$2,516,000	20% (\$498,000)
SHAW - WINEAGLE 138 kV	Rebuild line with higher-capacity conductor	\$4,842,000	100% (\$4,842,000)
FAP - COFFEYVILLE 138 kV	Reconductor line	\$1,149,000	100% (\$1,149,000)
TUCO - JOHN DAY SUBSTATION 138 kV	Reconductor line	\$665,000	23% (\$154,000)
	Total allocated to project		\$16,782,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.